

Experimental prototype of an open-source gas mixer for mechanical ventilation based on PID controllers in multiloop and parallel configuration.

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Abstract-- Objective: To design and implement a gas mixer for mechanical ventilators using a system of PID controllers in multiloop and parallel configuration. **Methods:** The design process included addressing the theoretical principles of gas mixing by partial volumes and their relationship with mechanical ventilation parameters. The electro-pneumatic system of the mixer, the control system, and its implementation were developed and documented in detail. **Results:** The system's performance was evaluated through experimental tests, and the results demonstrated effective control and reliable gas mixing according to the set parameters. **Conclusion:** The proposed design successfully achieves the intended gas mixing for mechanical ventilators, ensuring accurate and consistent gas delivery. **Significance:** This work provides a comprehensive solution, including all electronic designs, 3D components, and programming code, contributing to the advancement of mechanical ventilation technology.

Index Terms-- Mechanical ventilation, gas mixer, PID controllers, multiloop configuration, parallel control, experimental prototype.

I. INTRODUCTION

The gas mixer in a mechanical ventilator plays a crucial role in preparing the therapeutic mixture of air and oxygen that is delivered to the patient during treatment. Its primary function is to accurately adjust the oxygen volume fraction (FiO_2) in the inhaled gas mixture according to the specific needs of the patient. Additionally, the gas mixer actively participates in regulating the gas flow, taking into account parameters such as respiratory rate, airway pressure, and the set tidal volume. Precise control of these parameters is essential to ensure appropriate treatment, minimize the risks associated with mechanical ventilation, and ensure the successful implementation of the therapeutic strategy defined by the attending physician [1], [2], [3].

The technical characteristics of the gas mixer are defined based on the specifications of the mechanical ventilator and its performance. In 2020, the World Health Organization suggested the minimum requirements for mechanical ventilators used in the treatment of acute respiratory diseases such as COVID-19 [4]; among these requirements, it is

mentioned that an ICU mechanical ventilator should be able to regulate FiO_2 between 21% and 100%, produce inspiratory flows from 1 to 160 L/min, and be compatible with both high and low-pressure oxygen sources. These requirements directly apply to the gas mixing module of the ventilator, as it is responsible for managing these aspects in the mechanical ventilator.

Given the importance of the gas mixer in a mechanical ventilator, numerous studies have focused on its optimization. Soares et al. [5] conducted a comprehensive review of the different types of gas mixers used in mechanical ventilators, highlighting the implementation of Proportional-Integral-Derivative (PID) controllers to regulate the air and oxygen mixture. Taube [6] describes an air and oxygen mixing system that uses a proportional solenoid valve, controlled by a microprocessor to automatically adjust the FiO_2 according to the patient's blood oxygen saturation. Additionally, Taube [7] presents a system that not only mixes but also heats and humidifies the gases, with an adaptive PID control that adjusts the oxygen mixture delivered to the patient according to their needs. Radzi et. al [8] implemented a prototype that regulates the inspired oxygen fraction (FiO_2) using proportional flow control valves for air and oxygen, integrating sensors that allow dynamic adjustment of the gas flow.

While gas mixing systems that use PID controllers are a well-established technology, there is limited documentation on the use of multiloop controllers in this context. This study describes the design and implementation of an advanced gas mixing system aimed at applications in mechanical ventilation. A system has been developed that employs PID controllers organized in a multiloop and parallel configuration, designed to keep the control functions of different hardware aspects separate, under the premise that this separation of roles optimizes the overall system's precision and stability. In this regard, our development focuses on independently managing the individual flow of each gas through the implementation of cascaded control subsystems, where the output of one controller serves as the input for the next. This approach aligns with previous research that has demonstrated the effectiveness of

dual systems in mechanical ventilation, where different loops effectively manage various respiratory parameters [9].

This document is structured as follows: first, the theoretical foundations of gas mixing are described, defining the mathematical framework that underpins the prototype. Next, the conceptual design is presented, and the control schemes used are explained. Then, the implementation of the device is detailed, along with its performance metrics. All electronic designs, 3D components, and the complete program code are provided. Finally, an exhaustive analysis of the obtained results is offered.

II. THEORETICAL FOUNDATIONS OF GAS MIXING

Amagat's Law of Partial Volumes states that in a mixture of gases, the total volume of the mixture is equal to the sum of the individual volumes of each of the gases that make up the mixture, as if they were coexisting separately at the same temperature and pressure [10]. Mathematically, it is expressed as:

$$V_t = \sum_{i=1}^n V_i = V_1 + V_2 + \dots + V_n$$

$$1 = \sum_{i=1}^n \frac{V_i}{V_t} \quad (1)$$

where V_t corresponds to the total volume in the mixture of n gases. Consequently, the volume of each gas corresponds to a fraction of the total volume, such that their sum equals one.

In this context, air itself constitutes a mixture of different gases, with oxygen present at a concentration of 21% (typically at sea level), along with other gases such as nitrogen and argon, which together make up the remaining 79%. Applying equation (1), we can represent this mixture as follows:

$$V_M = 0.79V_M + 0.21V_M \quad (2)$$

where V_M is the total volume of the mixture, $0.21V_M$ is the total volume of oxygen, and $0.79V_M$ is the volume of the other components.

If one wished to prepare a mixture of oxygen-enriched air (air that receives an additional supply of oxygen) with a total volume V_M and an arbitrary oxygen concentration C_M defined in the range $0.21 \leq C_M \leq 1$, it would be understood that:

$$V_M = 0.79V_{air} + V_{to2} \quad (4)$$

where V_{air} is the volume of air added to the mixture and V_{to2} is the total volume of oxygen:

$$V_{to2} = 0.21V_{air} + V_{o2} \quad (5)$$

where V_{o2} is the volume of additional pure oxygen.

On the other hand, by applying equation (2), it can be deduced that the oxygen concentration C_M in the mixture V_M is given by:

$$C_M = \frac{V_{to2}}{V_M} \quad (6)$$

so that:

$$V_{to2} = C_M \cdot V_M \quad (7)$$

The concentration of the other gases would be given by:

$$0.79V_{air} = (1 - C_M)V_M$$

$$V_{air} = \frac{(1 - C_M)V_M}{0.79} \quad (8)$$

while the volume of oxygen V_{o2} :

$$V_{to2} = 0.21V_{air} + V_{o2} = C_M V_M$$

$$V_{o2} = C_M V_M - 0.21V_{air} \quad (9)$$

Directly measuring the volume can be impractical in this type of application, so we could redefine these formulas in terms of flow if we consider that V_M must be delivered during the inspiratory phase of ventilation, which has a duration T_i . Thus, we can define a target flow F_M as:

$$F_M = \frac{V_M}{T_i} \quad (10)$$

and consequently, the air flow F_{air} and oxygen flow F_{o2} required to generate the mixture:

$$F_{air} = \frac{(1 - C_M)F_M}{0.79} \quad (11)$$

$$F_{o2} = C_M F_M - 0.21F_{air} \quad (12)$$

Based on this, a gas mixer by partial volumes for a mechanical ventilator requires a method to control the flows of air and oxygen that are combined and delivered to the patient during the inspiratory phase of ventilation, and it must be able to repeat this process accurately with each breath.

III. PID CONTROLLER

The expression that defines the response of a PID (Proportional-Integral-Derivative) controller is determined by the following formula:

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt} \quad [11] \quad (13)$$

where K_p , K_i y K_d are the proportional, integral, and derivative coefficients of the system, respectively.

The function $e(t)$ is the error, defined as the difference between the desired value $m(t)$ and the measured value $l(t)$:

$$e(t) = m(t) - l(t) \quad (14)$$

On the other hand, applying the Laplace transform to equation (13), we obtain the transfer function:

$$\frac{U(s)}{E(s)} = K_p + \frac{K_i}{s} + K_d s \quad (15)$$

IV. CONCEPTUAL DESIGN

A gas mixer with cascaded PID controllers is proposed, where the output of the first controller serves as the input for the next internal controller. Figure 1 shows the functional model of the proposed gas mixer. The system consists of four (4) PID controllers in a multi-loop configuration, responsible for regulating the flow of each gas at an individual level: two PID controllers are located in the subsystem S_a (which controls the air supply) and two more in S_o (which controls the oxygen supply). These subsystems operate in parallel. The subscripts e , m , and l represent the error, the target value, and the measured value, respectively.

In figure 1, C_m y F_m are input parameters to the mixer coming from the ventilator, representing the target concentration and

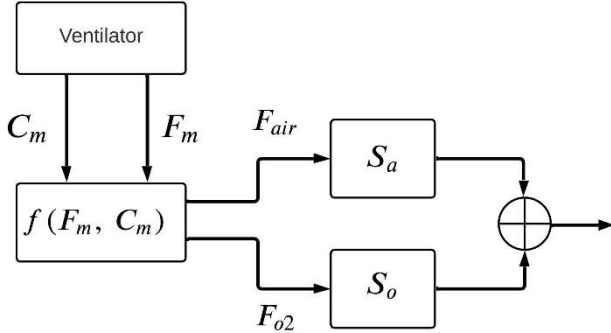


Fig. 1 – Functional model of the control system. S_a and S_o operate in parallel.

target flow, respectively. These parameters are used to determine the individual flows of air and oxygen using equations (11) and (12). This is represented by the block $f(F_m, C_m)$. Then, each resulting target flow value is processed separately in the identical subsystems S_a and S_o .

Figure 2 shows the internal composition of each subsystem. The input controller PID_f regulates the gas flow coming from the high-pressure source (air or oxygen). The parameter f_m corresponds to the target flow calculated in the block $f(F_m, C_m)$, meaning F_{air} or F_{o2} as applicable, and f_l to the measured flow.

The output I_m from PID_f serves as the input to the PID_i controller, whose purpose is to regulate the current in a proportional valve, which in turn determines the gas flow supplied. Since in proportional valves the flow intensity is proportional to the current, the objective of this subsystem is to keep the current close to a target value (I_m), which corresponds to the desired calculated flow. In this way, PID_i converts I_m into the signal O_i which is sent to the physical valve control system E_c , i.e., the electronic circuit, and finally to the valve.

By applying equation (13) to PID_f , the target current I_m that must be supplied to the valve is obtained:

$$I_m(t) = K_p f_e(t) + K_i \int f_e(t) dt + K_d \frac{df}{dt} \quad (16)$$

where f_e refers to the error or difference between the target flow f_m and the measured flow f_l . By applying equation (14) to f_e :

$$f_e(t) = f_m(t) - f_l(t) \quad (17)$$

Similarly, for PID_i we obtain that:

$$o_i(t) = K_p I_e(t) + K_i \int I_e(t) dt + I_d \frac{dI}{dt} \quad (18)$$

$$I_e(t) = I_m(t) - I_l(t) \quad (19)$$

where I_e corresponds to the current error, I_m to the target current, and I_l to the measured current.

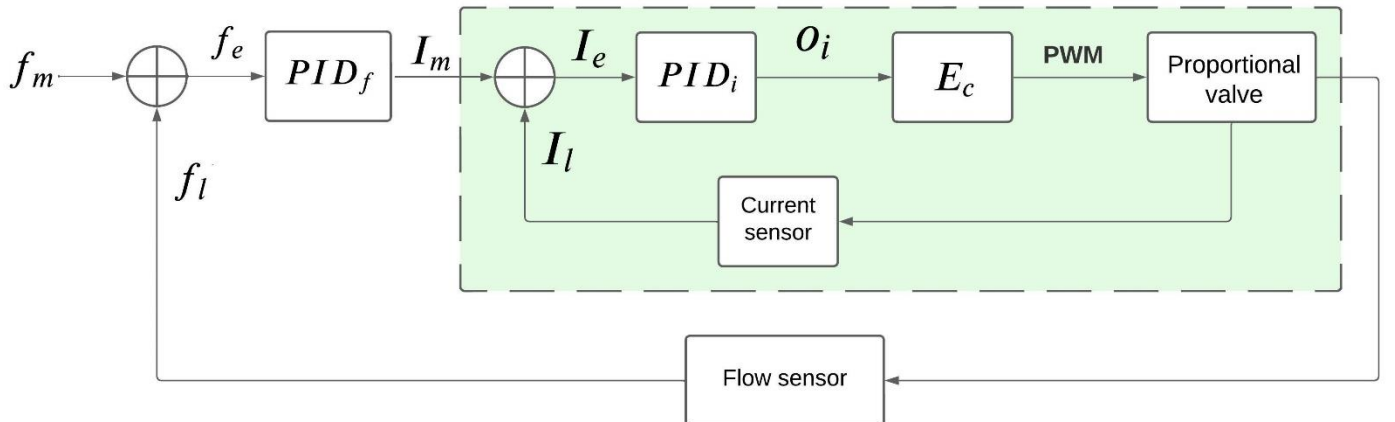


Fig. 2 – Multiloop PID Controller Subsystem for Air and Oxygen Control

Finally, the outputs of both subsystems are combined, directing the resulting gas flow to the patient, where the oxygen concentration is measured.

V. IMPLEMENTATION

Figure 3 shows the developed pneumatic system, and Figure 4 presents the electronic board designed according to the proposed design. The electronic schematics, software, blueprints, and drawings are available in the GitHub repository of IDEB for the Panavent project [12].

A. Pressure Regulators and Sensors

The objective of the regulators is to ensure that the working pressure remains within the tolerance limits of the flow sensors and proportional valves. According to the NFPA 99 standard, the output pressure in medical gas installations (such as oxygen, medical air, nitrogen, etc.) must be regulated to 50 psi, so the design was carried out considering this criterion.

The OxyReg model POR234KM1250 regulator was used. This device allows a maximum flow of 700 l/min, tolerates input pressures of up to 260 psi, and provides a stable output of 50 psi [13].

To monitor the pressure of the gases entering the system, two Honeywell SSCMANN006BGAA3 sensors [14] were used to provide feedback to the control system and generate alarms. This is crucial because the performance of certain devices, such

as proportional valves, depends on the operating pressure, which must be maintained within the limits defined by the manufacturer.

B. Input Flow Sensors

Two Sensirion SFM4200 [15] flow sensors were used, one for each gas. This is a digital sensor with I2C output, internally linearized, and with integrated temperature compensation. Its measurement range is from 0 to 160 slm, and it can operate at a maximum pressure of 8 bar (approximately 116 psi), with an accuracy of 2.5% to 5% of the measured value and a repeatability of 0.02% within its measurement range. This sensor was selected due to its performance characteristics in high-pressure applications.

The original structure of the sensor was partially modified. The gas inlet and outlet couplings were replaced with 3D-printed parts to adapt them to the manifold of the proportional valves on one side and to the gas inlets on the other. To prevent gas leaks, each coupling is fitted with a double rubber ring or O-ring. This is illustrated in figure 5.

C. Proportional valves

To control the flow of each gas, two proportional valves were selected. These valves consist of a coil wound around a metal plunger, designed in such a way that it can move in and out of its casing, operating the valve's opening and closing mechanism. Due to their inductive nature, when a voltage is applied across the coil terminals, the current flow begins to increase gradually, generating a magnetic field. Once the

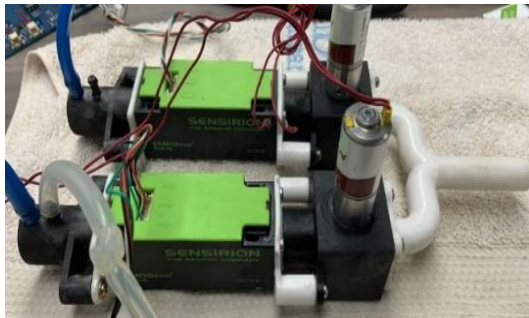


Fig. 3 - Pneumatic System.



Fig. 4 - Electronic System.

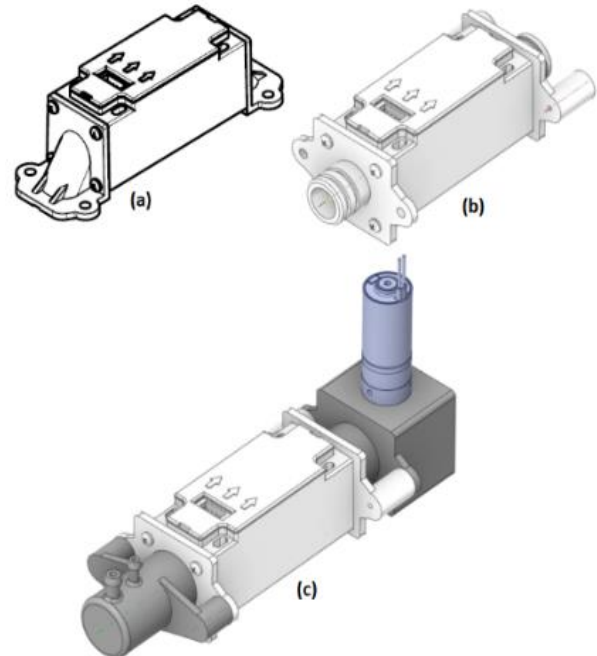


Fig. 5 – Flow Sensor Modifications: (a) the sensor as shown in the datasheet, (b) modifications made, (c) the sensor coupled to the valve manifold (right) and the inlet nozzle (left).

magnetic field reaches sufficient strength, an electromotive force is produced that displaces the plunger, allowing gas to flow through the valve. Therefore, the flow is regulated by controlling the current passing through the solenoid coil: the higher the current, the greater the gas flow, and vice versa [16].

As a tool for selecting the manufacturer and model of the valve, four evaluation criteria were defined:

1. Ability to supply the target flow F_M at the maximum and minimum extremes of V_t and T_i , evaluating equation (10.)
2. High repeatability in valve positioning, meaning reasonably low hysteresis.
3. Low power consumption.
4. Seat or seal compatible with oxygen.

The model DV-PM-10-600050-V valve [17], manufactured by Clippard, was selected. This valve offers a performance of 60 l/min at an input pressure of 50 psi, notable for its high linearity and very low hysteresis. The seal material is fluorinated rubber (FKM), suitable for handling medical oxygen according to ASTM International standard D1418 [18]. The valve was mounted on a 3D-printed manifold using biocompatible resin. The valve was secured to the manifold using a metal insert.

D. Static Mixer

After the proportional valves, an in-line static mixer was placed to homogenize the mixture. The mixer has a diameter of 5.5 mm, a length of 58 mm, and contains 3 blades, each 19.5 mm long. The blades are arranged in such a way that they alternate 180-degree turns to the right and left, so that the front end of one blade is perpendicular to the rear end of the next. The opposing direction of rotation for each blade maximizes the mixing effect [19].

E. Microcontroller and Firmware

The dsPIC33CK32MC102 microcontroller (μ C) [20] was used as the central processing unit, and it was programmed in C using MPLAB X IDE v6.05 [21].

The system clock was programmed to operate at a frequency of 120 MHz using an external 10 MHz MEMS resonator (DSC1001DL5-010.0000 [22]), utilizing the built-in PLL module of the microcontroller.

This code was modified to include a timeout routine in the I2C module, allowing the system to exit the wait loop in case of communication bus errors, such as when a slave device holds the SCL line low due to a malfunction, electrical fluctuations, or a poor connection. Additionally, a routine was implemented to forcibly recover the I2C bus in the event of a collision. Furthermore, drivers were developed to manage the shared use of timers 1 and 3, along with interfaces to handle various devices in the circuit, including I2C gates, external ADCs, and

Sensirion SFM flow sensors. All of this is integrated into the "panavent.hardware.drivers" library.

On the other hand, the logic of the gas mixer was implemented in the "panavent.mezclador" library. This is where the code that performs the calculations for preparing the mixture and reads all the sensors involved in the process is located. The library is designed in such a way that it can be incorporated as a reference or module into the main program.

The mixing process is executed by invoking the "mezclar" function from the "panavent.mezclador" library, which receives the target flow and target concentration defined by the user as input parameters. This function should be invoked in a loop with a period no less than 1 ms, the time required for all involved devices to respond appropriately, especially the flow sensors [15].

The source code for the PID controllers is located in the "panavent.utilities" library. The implemented algorithm incorporates an EMA filter to smooth fluctuations in the derivative component. Additionally, the controller's output is restricted to a configurable range to prevent overflow caused by integral accumulation.

F. Operation

The main program invokes the "mezclar" function with the target concentration and flows provided by the ventilator (user). With this data, equations (11) and (12) are applied to calculate F_{air} and F_{O_2} . After this, the subsystems S_a and S_o are executed, where equations (16) and (18) are calculated.

Subsequently, the microcontroller sends a PWM signal to each channel of the high-side power switch ITS4040D-EP-D (each channel is connected to a proportional valve) with a duty cycle proportional to equation (18). During the high part of the PWM signal, the valve's inductor accumulates energy and conducts current. When the PWM cycle switches to the low part, the switch opens the circuit, and the solenoid discharges through the freewheeling diodes (D02 and D03) and the current sensing resistor. In this way, an average current proportional to the duty cycle is applied to the inductor.

To measure the current flowing through the valves, a 0.1 Ohm resistor is used for each channel (R19 and R20). This resistor is deliberately low to minimize its impact on circuit performance, particularly on the solenoid's charging time. The ends of the resistor are connected to the differential inputs of the operational amplifier INA2290A3IDGKT (U1), so that the voltage drop is sensed and amplified. A filter was placed at the output of U1 to reduce the ripple caused by the PWM before sending it to the microcontroller, where the current is calculated by applying Ohm's law. This current serves as the input to PID_i , allowing it to calculate the appropriate output.

VI. PID CONTROLLER TUNING

The tuning of the controllers was carried out using the classic Ziegler-Nichols method to ensure optimal adjustment of the proportional, integral, and derivative parameters, applying the closed-loop response rule with no-overshoot (NO-OV) as detailed in [23]. The proportionality coefficients obtained for the controllers are shown in Table II.

The measurements were taken using a 200 MHz Siglent oscilloscope and a Fluke VT650 gas analyzer (Fig. 6), both of which had been calibrated within the last year.

To tune PID_i the voltage at the output of the current sensing circuit, i.e., at the input of the microcontroller's ADC, was measured. For tuning the other controllers, the flow at the output of the pneumatic system was measured.

VII. PID PERFORMANCE METRICS

To evaluate the performance of each PID controller, the following key indicators (performance metrics) were recorded in response to a step signal [6]:

- **Steady-State Value (SS):** This is the final value reached by the system's output, which remains constant after a period of time under constant input conditions.

TABLE II. PID PROPORTIONALITY COEFFICIENTS

Controller	K_p	K_i	K_d
PID_i	5.0	300.1852	0.022
PID_f	0.0015	0.025	0.00003

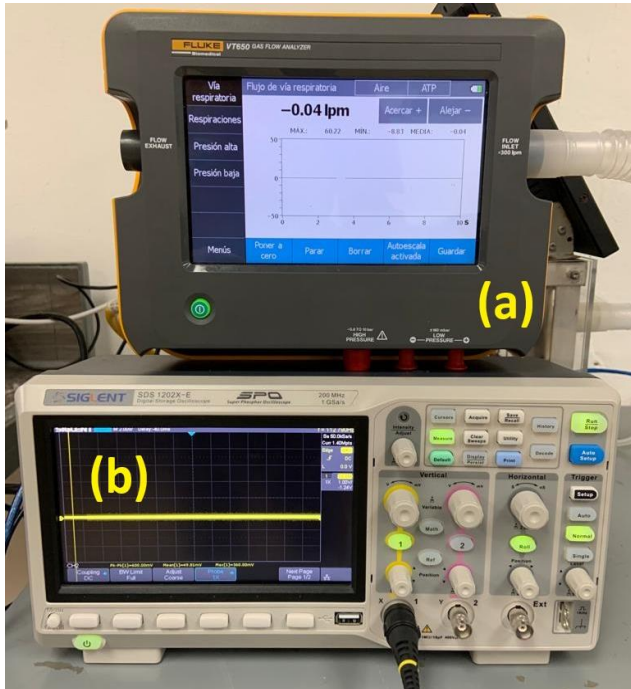


Fig. 6 – (a) Siglent 200MHz oscilloscope and (b) Fluke VT650 flow analyzer, used to make the measurements.

- **Percent Error:** The percentage difference between the steady-state value and the target value.
- **Rise Time (Tr):** The time it takes for the system's output to go from 10% to 90% of the steady-state value.
- **Settling Time (Ts):** The time it takes for the system's output to reach and stay within a $\pm 5\%$ band around the steady-state value.
- **Overshoot (Ov):** The difference between the maximum value reached by the output and the steady-state value. In tables where overshoot is not shown, its value was close to 0% of the final value, so it was omitted.

The procedure used for the calculation and measurement of each performance metric was as follows:

1. A target value (f_m or I_m) was defined and applied to the PID controller as one of its inputs.
2. The input was maintained for 2 seconds, during which the controller's output data was captured using one of the aforementioned measurement instruments. Then, the input was set to 0 for an additional 2 seconds.
3. This process was repeated 10 consecutive times (10 steps).
4. The measurements generated by the instrument were exported.
5. The key indicators for each step were calculated and averaged. This process was performed using the "Analyzer.cs" script [12], specifically designed for this purpose.
6. This process was repeated using a new target value.

Figure 7 shows the step response graph of the PID_i controller, and Table III details the performance metrics at different target current values I_m .

Figure 8 shows the step response graph of the flow controller PID_f and Table IV details the performance metrics at different target flow values f_m .

TABLA III. PID_i PERFORMANCE METRICS

I_m (A)	SS	% Error	Tr (ms)	Ts (ms)
0.10	0.098208	1.79	34.04	38.22
0.11	0.107781	2.02	33.27	38.89
0.12	0.118005	1.66	26.53	30.08
0.13	0.127449	1.96	25.14	29.34
0.14	0.137080	2.09	23.85	28.02
0.15	0.147032	1.98	22.64	26.22
0.16	0.156727	2.05	24.18	26.95
0.17	0.166653	1.97	22.28	26.18
0.18	0.176432	1.98	18.05	19.42
0.19	0.186067	2.07	16.03	21.62

TABLE IV. PID_f PERFORMANCE METRICS

f_m (l/min)	SS	Ov.	% Error	Tr (ms)	Ts (ms)
10.00	9.94	0.29	0.61	253.09	355.75
15.00	14.94	0.34	0.37	369.85	513.64
20.00	19.92	0.45	0.41	406.07	550.50
25.00	24.94	0.67	0.23	430.57	562.03
30.00	29.87	0.69	0.45	437.67	590.18
35.00	34.83	0.86	0.49	469.86	653.38
40.00	39.75	0.71	0.61	536.37	749.13

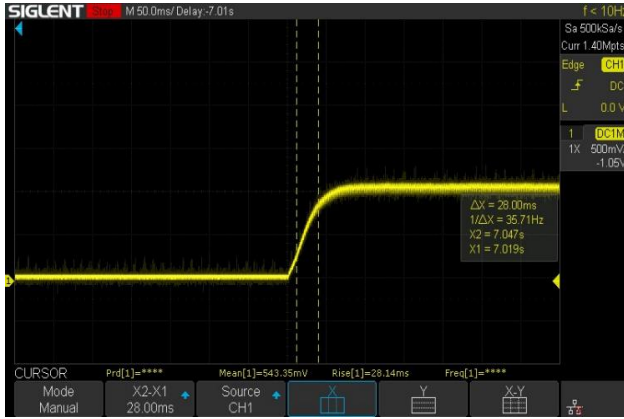


Fig. 7 – PID_i step response.

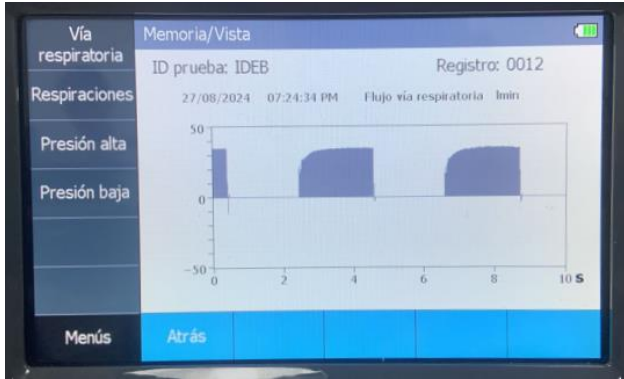


Fig. 8 – PID_f step response.

VIII. GAS MIXER PERFORMANCE METRICS

To evaluate the efficiency of the system as a whole, oxygen concentration measurements were taken under different operating conditions. The results obtained are presented in Table V. The procedure followed for these measurements was as follows:

1. The gas analyzer VT650 was connected to the system's output.

2. A target flow value F_m (starting with 10 L/min) and a target concentration value C_m (starting with 25%) were set.
3. A 20-second wait period was observed to allow the oxygen sensor of the analyzer to reach equilibrium.
4. The measured oxygen concentration was recorded.
5. Steps 2-4 were repeated with different combinations of F_m and C_m .

IX. RESULTS AND DISCUSSION

The PID_i controller shows consistent performance across the entire operating range, with low percentage errors and fast response times:

- The steady-state value improves as the target current increases, which is especially beneficial when controlling a high gas flow.
- The percentage error varies slightly between 1.66% and 2.09%, with a mean deviation of 0.093% and a standard deviation of 0.127%. These are relatively low values, indicating that the controller is quite accurate. Additionally, the low dispersion of percentage errors around the mean suggests that the PID_i controller has uniform performance across its entire operating range.
- The rise time (Tr) decreases as I_m increases, from 34.04 ms to 16.03 ms. This behavior is favorable, as a shorter rise time means the system reaches the target value more quickly. Moreover, this indicates that the controller becomes more responsive at higher currents, which is advantageous in situations requiring a rapid response.
- Similar to Tr , the settling time (Ts) decreases as I_m increases, indicating better stability and quicker achievement of the steady state, which is indicative of a system that stabilizes relatively quickly after a change in input.

Regarding the PID_f controller, it stands out for its good accuracy in tracking the target flow, maintaining low percentage errors throughout its entire operating range. However, a slight decrease in performance is observed as the flow increases:

- The percentage error slightly decreases as f_m increases, reaching its lowest value (0.23%) at 25 L/min. After this point, the percentage error slightly increases again, reaching 0.61% at 40 L/min. Overall, the percentage errors remain low, with a reduced mean deviation (0.1%) and standard deviation (0.12%), indicating that the controller maintains high accuracy and minimal variability in flow regulation across its entire operating range.
- The overshoot increases with the rise in f_m , going from 0.29% to 0.86%. Although the overshoot values remain relatively low, this increase indicates that the system tends to slightly exceed the target value before stabilizing, especially at higher flows.

- The rise time and settling time consistently increase with the rise in f_m . This suggests that the system requires more time to reach 90% of the steady-state value at higher flows.

On the other hand, the overall performance metrics of the system indicate that the gas mixer operates with acceptable efficiency and precision:

- The average concentration calculated for each combination of target flow F_m and target concentration C_m is quite close to the target value. This suggests that the gas mixer is operating with high accuracy under most of the evaluated conditions.
- The average concentrations align well with the target concentrations, especially in higher flow and concentration ranges, which is a good indication of the system's stability and reliability.
- The absolute error ranges from 0.40 to 1.06, indicating a good level of precision in gas mixing. The highest errors are observed at lower C_m concentrations, which may suggest that the system is slightly less accurate in these ranges. However, even the highest absolute error of 1.06 is still relatively low, indicating that the gas mixer maintains overall good precision.
- The percentage error ranges from 0.44% to 4.23%, being higher at lower target concentrations (25.00%) and decreasing as the target concentration increases. This behavior suggests that the system is more efficient in controlling mixtures with higher oxygen concentrations. The generally low percentage error reinforces the idea that the system is well-tuned and capable of providing

accurate gas mixtures under a variety of operating conditions.

- The standard deviation ranges from 0.08 to 0.48, indicating variability in the mixer's precision depending on the target flow and concentration. Higher standard deviations in high flows and elevated concentrations might suggest slight fluctuations in system performance under these conditions. However, most standard deviations are quite low, indicating reasonable consistency in the system's precision.

X. CONCLUSIONS

The analysis of the system reveals a robust and efficient design, suitable for applications in mechanical ventilation. The multiloop PID controllers, both for current and flow, demonstrated high performance in step response, with low percentage errors and minimal variability.

Additionally, the gas mixer performance metrics, measured through oxygen concentration under various conditions, confirm that the system maintains high precision in regulating mixtures. This is evidenced by the overall operation of the system with reduced standard deviation, indicating consistent and reliable performance.

Overall, the designed gas mixing system proves to be efficient and precise. However, optimizing performance under extreme flow and concentration conditions could further enhance the system's reliability in high-demand situations.

TABLA V. GAS MIXER PERFORMANCE METRICS

		Target flow F_m (L/min)							Avg.	Error	% Error	σ
		10	15	20	25	30	35	40				
Target concentration C_m (%)	25.00	26.10	26.10	26.10	26.00	26.10	26.10	25.90	26.06	1.06	4.23	0.08
	30.00	30.40	30.80	31.10	30.80	30.90	30.90	31.00	30.84	0.84	2.81	0.22
	35.00	35.90	35.80	35.90	35.70	35.70	35.90	35.90	35.83	0.83	2.37	0.10
	40.00	40.10	40.70	40.80	40.80	40.70	40.80	40.80	40.67	0.67	1.68	0.26
	45.00	45.30	45.80	45.80	45.70	45.50	45.50	45.70	45.61	0.61	1.37	0.19
	50.00	50.10	50.70	50.60	50.60	50.50	50.40	50.50	50.49	0.49	0.97	0.20
	55.00	55.20	55.60	55.60	55.60	55.40	55.40	55.20	55.43	0.43	0.78	0.18
	60.00	60.10	60.50	60.60	60.60	60.50	60.40	60.20	60.41	0.41	0.69	0.20
	65.00	65.20	65.40	65.60	65.70	65.40	65.40	65.20	65.41	0.41	0.64	0.19
	70.00	70.20	70.20	70.50	70.60	70.50	70.40	70.20	70.37	0.37	0.53	0.17
	75.00	75.90	75.40	75.40	75.40	75.40	75.30	75.10	75.41	0.41	0.55	0.24
	80.00	81.10	80.60	80.50	80.10	80.20	80.10	80.10	80.39	0.39	0.48	0.38
	85.00	86.10	86.10	85.80	85.80	85.50	85.00	85.90	85.74	0.74	0.87	0.39
	90.00	91.10	90.80	90.60	90.40	90.10	90.10	89.70	90.40	0.40	0.44	0.48
	95.00	95.70	95.80	95.70	95.40	95.20	95.20	95.10	95.44	0.44	0.47	0.29

XI. ACKNOWLEDGMENTS

This work is a derivative research project from Panavent (COVID19-P-228), funded by the National Secretariat of Science and Technology (SENACYT) of the Republic of Panama, through the emergency funding program for research projects during the COVID-19 pandemic.

XII. REFERENCES

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